Phase King Algorithm

(variables)
boolean: $v \longleftarrow$ initial value;
integer: $f \longleftarrow$ maximum number of malicious processes, $f < \lceil n/4 \rceil$;

(1) Each process executes the following $f+1$ phases, where $f < n/4$:
(1a) **for** *phase* $= 1$ **to** $f+1$ **do**
(1b) Execute the following round 1 actions:
(1c) **broadcast** v to all processes;
(1d) **await** value v_j from each process P_j;
(1e) *majority* $\longleftarrow$ the value among the v_j that occurs $> n/2$ times (default value if no majority);
(1f) *mult* $\longleftarrow$ number of times that *majority* occurs;
(1g) Execute the following round 2 actions:
(1h) **if** $i = phase$ **then**
(1i) **Pi broadcast** its *majority* to all processes;
(1j) **receive** *tiebreaker* from P_{phase} (default value if nothing is received);
(1k) **if** $mult > n/2 + f$ **then**
(1l) $v \longleftarrow majority$;
(1m) **else** $v \longleftarrow tiebreaker$;
(1n) **if** $phase = f+1$ **then**
(1o) output decision value v.

Algorithm 14.4 Phase-king algorithm [4] – polynomial number of unsigned messages, $n > 4f$. Code is for process P_i, $1 \le i \le n$.

If neither 0 or 1's are greater than n/2,
which may happen when the malicious processes do not respond,
and the correct processes are split
among themselves, then a default
value is used for the majority variable.

Proof

There are f+1 phases but only f malicious process hence at least one of the phase kings is non malicious. Let that be for phase K.

Consider the three cases that might be true for two non malicious process Pi and Pj.

1. Pi and Pj decided on their majority value at the end of round K
2. Pi and Pj decided on the phase king value at the end of round K
3. Pi decided on majority and Pj decided on phase king value at the end of round K

Proof

As the phase king of phase k is non-malicious, all non-malicious processes can be seen to have the same estimate value v at the end of phase k. Specifically, observe that any two non-malicious processes P_i and P_j can set their estimate v in three ways:

(a) Both P_i and P_j use their own *majority* values. Assume P_i's *majority* value is x, which implies that P_i's $mult > n/2 + f$, and of these voters, at least $n/2$ are non-malicious. This implies that P_j must also have received at least $n/2$ votes for x, implying that its majority value *majority* must also be x.

(b) Both P_i and P_j use the phase king's tie-breaker value. As P_k is non-malicious it must have sent the same tie-breaker value to both P_i and P_j.

(c) P_i uses its majority value as the new estimate and P_j uses the phase king's tie-breaker as the new estimate. Assume P_i's *majority* value is x, which implies that P_i's $mult > n/2 + f$, and of these voters, at least $n/2$ are non-malicious. This implies that phase king P_k must also have received at least $n/2$ votes for x, implying that its majority value *majority* that it sends as tie-breaker must also be x.

Proof

Once a non malicious phaseking executes his two rounds; the values of the non malicious processes will not change with any further phase king rounds

Going forward, all the non faulty processes already have agreed on a same value. That is more than 3n/4 process now have the same value of v which they send everyone else in round1 of future phases. So all non faulty ppl will get a clear majority of > n/2+f and will continue to decide on their v value.